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Paper 1: New Frontiers of Drug Development Through the Use of New Approach Methodologies

Bibliographic Information

- **Title:** New Frontiers of Drug Development Through the Use of New Approach Methodologies
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Paper Type: GENERIC

Quick Take

This meeting report summarizes a November 2025 AAPS PharmSci 360 symposium on New Approach Methodologies (NAMs) for drug development. It covers the qualification of microphysiological systems (MPS) through a tissue chip testing consortium, the use of 3D organoids (including patient-derived and canine models) for safety, efficacy, and personalized medicine, and the growing utility of in silico tools (PBPK, AI/ML) to reduce reliance on animal testing.

Executive Summary

This report synthesizes expert presentations from an AAPS symposium on NAMs, focusing on three pillars: MPS, 3D organoids, and computational tools. The MPS section details a multi-sector consortium's work on context-of-use qualification and comparative evaluations for safety assessments. The organoid discussion highlights their potential in personalized medicine, infectious disease modeling, and comparative oncology (including canine organoids). The in silico section emphasizes how PBPK modeling, artificial intelligence, and machine learning can integrate diverse data sources to predict toxicity and pharmacokinetics faster than traditional methods. The report aligns with FDA/EMA initiatives to modernize preclinical safety testing and reduce animal use.

Scientific Context & Motivation

The paper addresses the critical need to transition from traditional animal-based preclinical testing toward human-relevant, efficient New Approach Methodologies (NAMs). Key knowledge gaps include the lack of standardized qualification frameworks for MPS, limited understanding of organoid reproducibility and scalability, and the need for robust validation of in silico models for regulatory decision-making. The symposium and this report aim to share current experiences and catalyze broader adoption of NAMs across academia, industry, and regulatory agencies.

✂ Methodological Snapshot

The symposium covered three methodological domains: (1) Microphysiological Systems (MPS) — evaluated through the TEX-VAL tissue chip testing consortium, which performs context-of-use qualification and inter-laboratory comparisons across platforms like liver acinus chips, kidney proximal tubule chips, and intestinal barrier models. (2) 3D Organoids — derived from adult stem cells (e.g., intestinal, hepatic, tumor organoids) and cultured in Matrigel or synthetic hydrogels; they can be co-cultured with immune cells or vascularized and are used for drug screening, permeability studies, and disease modeling. (3) In Silico Tools — including PBPK models for IVIVE, machine learning algorithms for toxicity prediction, and AI-driven integration of multi-omics data; these tools reduce reliance on predefined models and enable rapid hypothesis generation. The presenters highlighted the need for fit-for-purpose validation and regulatory engagement for each approach.

📋 Detailed Analysis

The report is structured around three thematic sessions. In the MPS session, Rusyn et al. described the TEX-VAL consortium's systematic approach: sourcing multiple MPS platforms, running standardized test compounds, and comparing results across labs. A key point is that reproducibility was acceptable for some endpoints (e.g., hepatotoxicity biomarkers) but variable for others (e.g., transporter-mediated clearance). The organoid session, led by Mochel and Allenspach, emphasized the translational value of canine organoids — given the outbred nature and spontaneous disease in dogs, these models may predict human responses better than inbred rodent models. They also discussed technical hurdles such as Matrigel variability and the need for defined media. The in silico session showcased how AI/ML can mine literature and public databases to flag toxicity risks and how PBPK models can incorporate organoid-derived clearance data. A notable example was the combination of kidney MPS data with PBPK to predict renal clearance of PFAS and drugs. The report aligns with the FDA's 2025 roadmap to reduce animal testing and the EMA's reflection paper on AI. However, as a meeting report, it lacks critical depth on statistical validation, uncertainty quantification, or comparison of different modeling approaches. The graphical abstract is well-designed to convey the synergies among the three pillars.

Domain Context

This report sits at the intersection of regulatory science, preclinical toxicology, and model-informed drug development (MIDD). It reflects a paradigm shift where regulators (FDA, EMA) are proactively developing frameworks for NAMs, mirroring earlier efforts to qualify in vitro assays and PBPK modeling. The pharmacometrics community will find the in silico section most directly relevant, particularly the integration of MPS/organoid data into PBPK models for IVIVE. The symposium also underscores the importance of consortia (like IQ MPS Affiliate and TEX-VAL) in generating the evidence base needed for regulatory endorsement. This collaborative model is analogous to how the MIDD community advanced PBPK qualification through cross-industry working groups. The report implicitly calls for quantitative scientists to contribute to NAMs

validation by developing statistical frameworks for reproducibility, predictive performance, and uncertainty assessment.

☒ Key Findings

- 1) Microphysiological systems (MPS) are being systematically evaluated through the TEX-VAL tissue chip testing consortium, which performs context-of-use testing and comparative evaluations to support regulatory acceptance in safety assessments.
 - 2) 3D organoids, including patient-derived tumor organoids and canine intestinal organoids, offer promising platforms for personalized medicine, drug permeability studies, and comparative oncology, but require standardization and validation.
 - 3) In silico tools—PBPK models, AI, and ML—enable rapid integration of diverse data to predict drug toxicity, population variability, and pharmacokinetics, with some models generating predictions in hours versus weeks for traditional methods.
 - 4) Regulatory bodies (FDA, EMA) are actively developing guidance and accepting NAMs submissions, as evidenced by recent FDA documents and the ‘Roadmap to Reducing Animal Testing.’
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Strengths & Limitations

Strengths

- Comprehensive overview of three major NAMs pillars (MPS, organoids, in silico) with expert perspectives from academia, industry, and government.
- Includes practical consortium experiences (TEX-VAL) that demonstrate systematic qualification approaches for MPS.
- Highlights emerging areas such as canine organoids for comparative oncology and AI/ML integration with PBPK.
- Places NAMs within the current regulatory landscape by citing FDA/EMA guidance and initiatives.
- Balanced discussion of opportunities and challenges (e.g., reproducibility, scalability, context-of-use requirements).

Limitations (Acknowledged by Authors)

- The report is a meeting summary and does not present new experimental data or direct comparisons of NAMs performance.
- Authors note the need for further standardization, qualification, and regulatory guidance for widespread adoption.

Limitations (Expert Review)

- Depth of coverage is limited by the symposium format; each NAMs category is described broadly without rigorous quantitative analysis.
- Lack of direct evidence comparing NAMs predictive performance against traditional animal models or clinical outcomes.
- Minimal discussion on cost, infrastructure requirements, or training barriers that may hinder adoption.

- No mention of potential biases in AI/ML models or strategies to mitigate them beyond a brief reference.

Generalizability

The concepts and frameworks discussed (context-of-use qualification, consortium-driven validation) are broadly applicable across therapeutic areas and drug modalities. However, the specific examples focus on hepatic, renal, intestinal, and tumor models, so direct generalizability to other tissues or complex systemic interactions may require further work. The regulatory perspectives are US- and EU-centric.

Figures & Tables

- **Figure 1:** Graphical abstract summarizing the three categories of NAMs discussed: Microphysiological Systems (MPS), 3D Organoids, and In Silico Tools (PBPK, AI/ML). Icons represent tissue chips, organoid cultures, and computer models.
 - *Significance:* Provides a visual overview of the symposium's scope and the interconnections among the three NAMs pillars. Emphasizes the complementary nature of these approaches in replacing or reducing animal testing.
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Future Directions

Future work should focus on: (1) developing standardized validation protocols and performance metrics for MPS and organoids across multiple laboratories; (2) integrating MPS/organoid data with PBPK and AI/ML models for quantitative prediction of human pharmacokinetics and toxicity; (3) expanding the library of context-of-use qualifications to cover more organ systems and disease states; (4) conducting prospective studies that compare NAMs-based predictions to clinical outcomes; (5) addressing scalability and cost barriers for routine industrial use; and (6) harmonizing regulatory acceptance criteria globally.

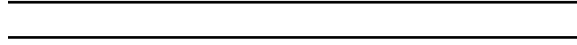
Expert Commentary

This symposium report captures a pivotal moment in preclinical drug development. The systematic qualification of MPS through consortia is a pragmatic path to regulatory acceptance, reminiscent of the early days of PBPK modeling. The inclusion of canine organoids for comparative oncology is a particularly innovative angle that bridges veterinary and human medicine. For pharmacometricians, the integration of in silico tools with experimental NAMs data represents a natural extension of MIDD principles. The report would benefit from more quantitative comparisons, but as a high-level roadmap, it effectively conveys the momentum behind NAMs.

Bottom Line

NAMs are transitioning from research tools to regulatory-relevant platforms. Practitioners in drug development and pharmacometrics should actively engage with consortium efforts (e.g., TEX-VAL), familiarize themselves with evolving FDA/EMA guidance on NAMs, and consider

integrating MPS, organoids, and in silico modeling into their own workflows to enhance human relevance and reduce animal use. The report underscores that context-of-use validation is the key to regulatory acceptance.



☒ **Figures**

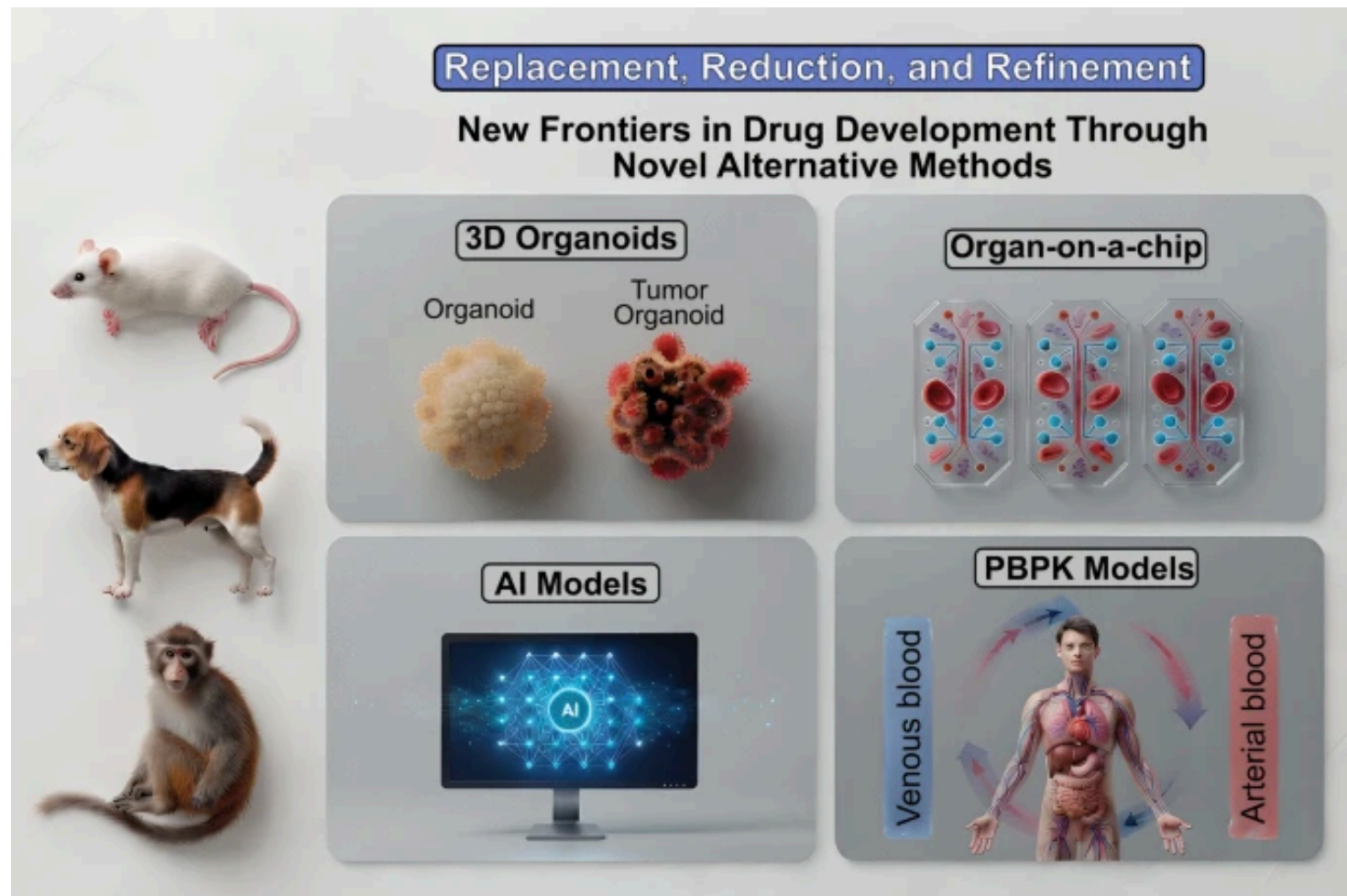


Figure 1: Figure 1